

Q.6a (20 marks): The hallmark of man is bipedalism. Discuss how bipedalism may have evolved and state its evol...

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Q6. (a) The hallmark of man is bipedalism. Discuss how bipedalism may have evolved and state its evolutionary implications

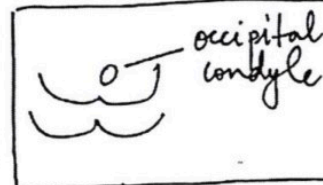
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About 6 million years ago, most significant thing happened in human evolution that is erect posture and bipedalism.

Evolution

1) Skull

- presence of mastoid process.
- foramen magnum forward and downwards.
- The weight of the skull went behind body, there was less need of robust neck muscles to hold weight.



2) Spine

- presence of doubly elongated vertebral column.
 - ↳ curved
 - ↳ Thoracic
 - ↳ lumbar
 - ↳ sacral
- Thoracic and lumbar curve helped put center of ~~body~~ gravity of body weight in

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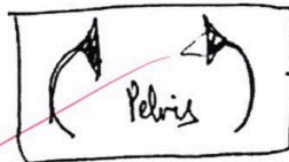
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one line enabling bipedalism.

3) Pelvis

- Pelvis become more ~~to~~ shorter and broader thus it helped to support internal organs and weight ~~stabil~~ stabilisation.



→ Fig: Human Pelvis

4) Hands

- Hindlimbs became longer than forelimbs. Thus two hands of man became free.

5) Feet

- Due to arched feet, more stability came to human body and less energy was ~~to~~ now required to walk.

reduce them & state
How it has evolved.

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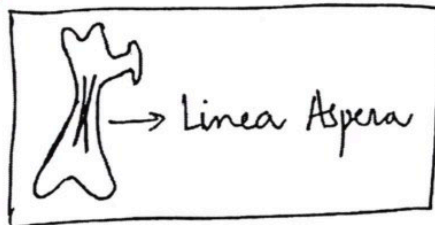
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→ try to know about
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6.) Femur

- presence of linea aspera in femur allowed development of extensor muscles which provided stability to erect posture and enabled bipedalism

Fig: FemurImplications(i.) Biological

- a) Bipedalism & erect posture allowed more effective walking than knuckle walking (McHenry 1980)
- b) More heat dissipation as lower body part exposed to environment (Pete Wheeler)

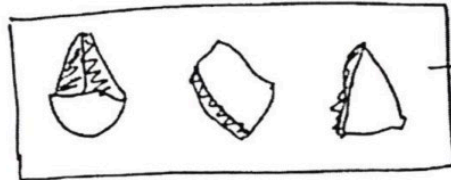
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- (C) Painful birthing due to lowering of birth canal.
- (D) Lower back problems due to more weight.

ii) Cultural

- (A) Enhanced social bonding ~~living~~ ^{living} in of mother-child.
- (B) Both hands were freed which allowed man to make tools and become a specialised hunter.



Tools paved way for culture.

Hence, bipedalism was a turning point in human evolution which ~~has~~ ^{has} ~~is~~ ^{gave} rise to ~~a~~ ^{an} advanced species.